

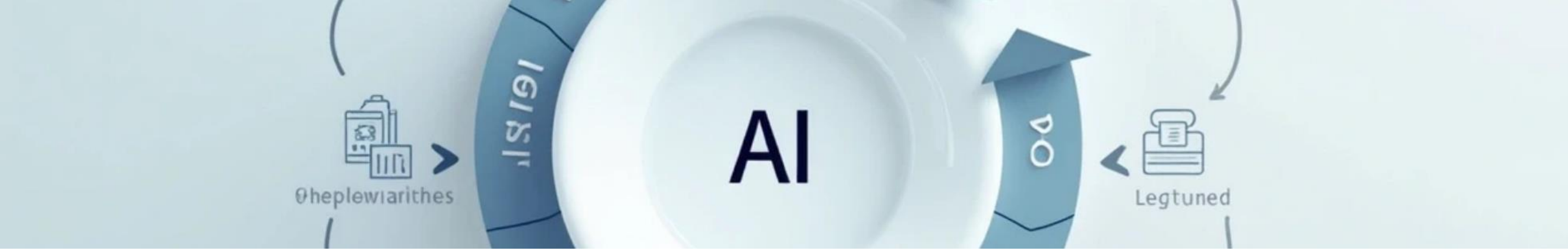
AI Fundamentals and Project Documentation



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01

Question 1: AI Fundamentals and Project Scoping (15 Marks)



AI Foundations (5 Marks)

Key difference between AI and ML (2 Marks)

Artificial Intelligence (AI) is the broad science of making machines act like humans in tasks such as reasoning, problem solving, or decision-making. Machine Learning (ML) is a branch of AI that allows systems to learn from data and improve automatically without being explicitly programmed. In short: AI is the goal, ML is one way to achieve it (Russell & Norvig, 2021).



AI Foundations (5 Marks)

Importance of historical overview of AI (1 Mark)

The history of AI helps new developers understand how ideas and technologies evolved over time. It shows both successes and failures, which prevents repeating mistakes and inspires better innovation (McCorduck, 2004).



AI Foundations (5 Marks)

Importance of AI in electronic devices + example (2 Marks)

AI makes devices smarter, faster, and more user-friendly. For example, in the healthcare industry, smart wearables like fitness watches use AI to track heart rates and detect irregularities, helping to prevent illnesses early. This improves health and reduces medical costs (Esteva et al., 2019).



Project Documentation (10 Marks)

Social, Economical, and Technical factors (3 Marks)

1

Social factor

The Smart Surveillance System improves community safety by reducing crime and false alarms (e.g., distinguishing pets from intruders).

2

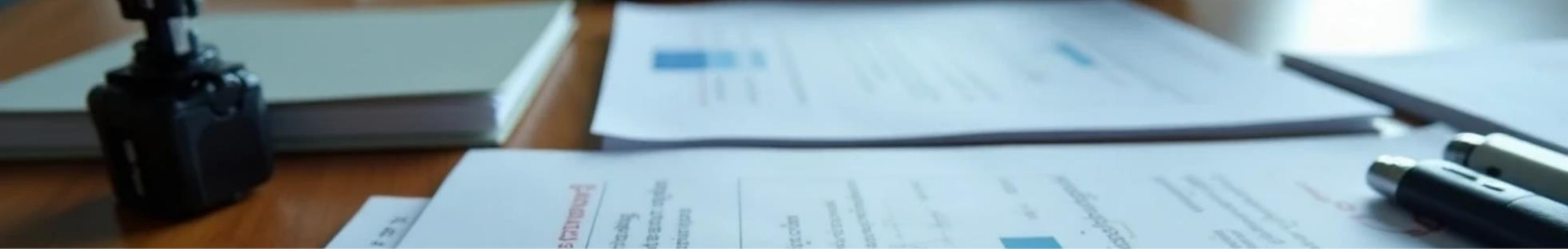
Economical factor

It lowers costs by reducing the need for multiple human security guards, making security more affordable.

3

Technical challenge

Ensuring accurate detection in low-light conditions or when pets resemble humans remains a challenge.



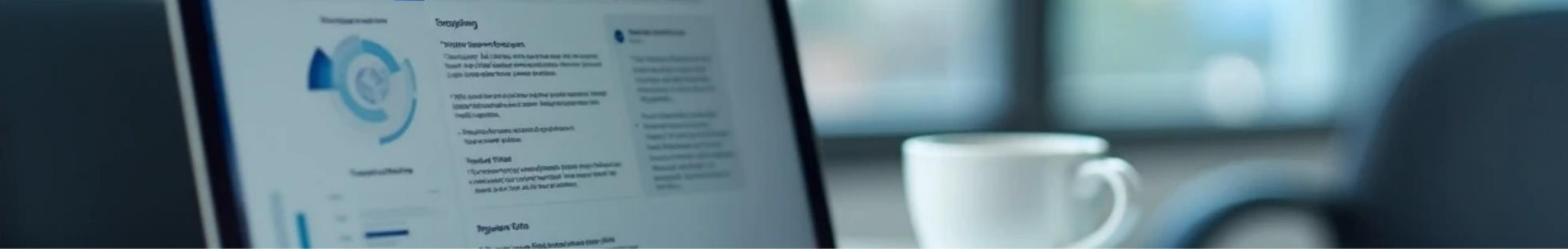
Project Documentation (10 Marks)

Three critical sources of information (3 Marks)

Research papers and journals on AI and computer vision (for the latest techniques).

Hardware manuals and camera datasheets (to understand device limits).

Government regulations/industry standards (to ensure privacy and ethical compliance).



Project Documentation (10 Marks)

Core function of documentation + two key elements (4 Marks)

The AI documentation process provides a roadmap for the project, ensuring clarity, collaboration, and accountability. It records essential details that guide development and future maintenance. Two key elements at this stage are:

1	2
Project goals and scope – what the AI system is meant to achieve.	System requirements – including data type, tools, and AI models to be used.

References

Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach. Pearson.

McCorduck, P. (2004). Machines Who Think. A K Peters.

Esteva, A., et al. (2019). "A guide to deep learning in healthcare." Nature Medicine, 25(1), 24–29.

Thank You